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BANGALORE | DELHI NCR | PUNE

Regression Analysis Lab

Course Code: MST511-2

Sales prediction

LAB: 04

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1.INTRODUCTION

Multiple Linear Regression is a statistical technique used to examine the relationship between a dependent variable and two or more independent variables. It is useful for understanding how several factors simultaneously influence an outcome and for developing models that can be used for prediction. In this analysis, **Sales** is considered the dependent variable, while **Advertising Costs, Store Size, Percentage of Employee Retention, Customer Satisfaction Score, and Promotion** are considered the independent variables.

The objective of this study is to develop an appropriate multiple linear regression model for predicting sales across 41 stores. The analysis aims to identify the variables that contribute significantly to sales and to determine the best combination of predictors for the regression model.

To select the most suitable set of explanatory variables, three variable-selection procedures—**Forward Selection, Backward Elimination, and Stepwise Selection**—are applied. The models obtained from these procedures are compared using appropriate statistical criteria to determine the best-fitting model. The final model can provide useful insights into the factors affecting store sales and can support better decision-making regarding advertising, store operations, employee retention, customer satisfaction, and promotional activities.

2.ANALYSIS

```
library(readxl)
data <- read_xlsx("C:/Users/raghu/OneDrive/Desktop/STUDIES/Guithub + Practica
l college/practicals/R programming/lab_4/store_sales_data.xlsx")
head(data)
```

```
## # A tibble: 6 × 7
##   Store `Advertising Costs` `Size (Sq. Ft)` `% Employee Retention`
##   <dbl>               <dbl>           <dbl>                   <dbl>
## 1     1                124.           22560                      59
## 2     2                154            31181                      62
## 3     3                124.           16314                      78
## 4     4                163            24205                      66
## 5     5                107            17574                      82
## 6     6                144.           19584                      67
## # ℹ 3 more variables: `Customer Satisfaction` <dbl>, Promotion <dbl>,
## #   Sales <dbl>
```

```
# dimension
dim(data)
```

```
## [1] 41  7
```

```
# structure of data
str(data)
```

```
## tibble [41 × 7] (S3: tbl_df/tbl/data.frame)
##  $ Store                : num [1:41] 1 2 3 4 5 6 7 8 9 10 ...
##  $ Advertising Costs    : num [1:41] 124 154 124 163 107 ...
##  $ Size (Sq. Ft)        : num [1:41] 22560 31181 16314 24205 17574 ...
##  $ % Employee Retention : num [1:41] 59 62 78 66 82 67 57 64 88 87 ...
##  $ Customer Satisfaction: num [1:41] 32 33 28 43 22 34 32 23 25 17 ...
##  $ Promotion            : num [1:41] 1 1 0 0 1 0 1 1 0 0 ...
##  $ Sales                : num [1:41] 1581 2139 1043 1702 1339 ...
```

2.1 - CORRELATION

```
library(Hmisc)

##
## Attaching package: 'Hmisc'

## The following objects are masked from 'package:base':
##
##     format.pval, units

data_corr <- data[,c("Store","Advertising Costs","Size (Sq. Ft)","% Employee
Retention","Customer Satisfaction","Promotion","Sales")]
corr_result <- rcorr(as.matrix(data_corr))
# correlation coefficients
corr_result$r

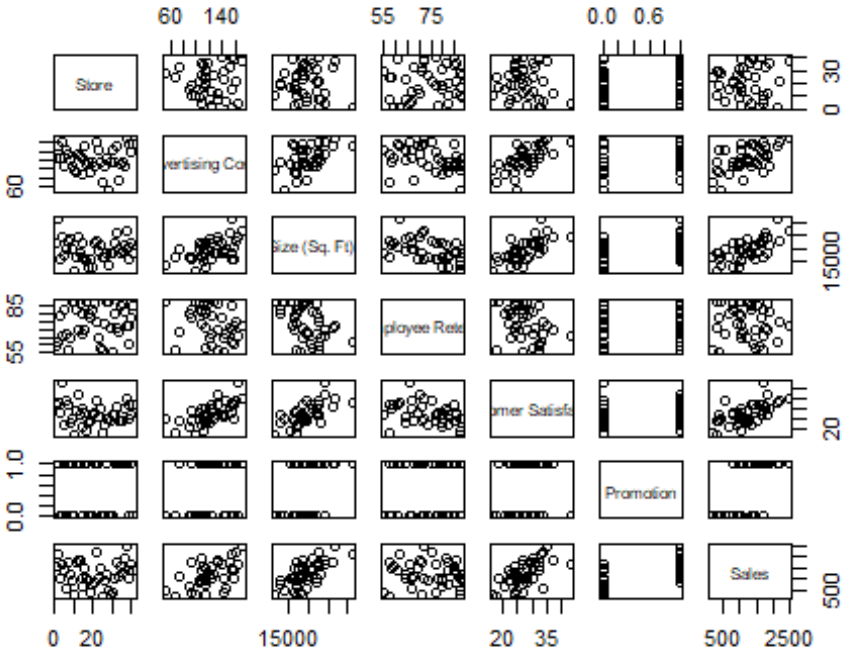
##                             Store Advertising Costs Size (Sq. Ft)
## Store                  1.000000000        0.06286544   -0.04894671
## Advertising Costs      0.062865439        1.00000000    0.51444877
## Size (Sq. Ft)         -0.048946710        0.51444877    1.00000000
## % Employee Retention   0.129647911       -0.31589450   -0.43201600
## Customer Satisfaction -0.119792782        0.65639254    0.66977533
## Promotion              0.004133785        0.17566707    0.46232196
## Sales                  0.011180465        0.58468934    0.66877440
##                       % Employee Retention Customer Satisfaction   Promoti
on
## Store                          0.12964791            -0.1197928 0.0041337
85
## Advertising Costs             -0.31589450             0.6563925 0.1756670
73
## Size (Sq. Ft)                 -0.43201600             0.6697753 0.4623219
64
## % Employee Retention           1.00000000            -0.3542745 0.0883067
62
## Customer Satisfaction         -0.35427448             1.0000000 0.1297613
07
## Promotion                      0.08830676             0.1297613 1.0000000
00
## Sales                         -0.08233431             0.6318326 0.6347649
17
##                              Sales
```

```
## Store                   0.01118046
## Advertising Costs       0.58468934
## Size (Sq. Ft)           0.66877440
## % Employee Retention   -0.08233431
## Customer Satisfaction   0.63183257
## Promotion               0.63476492
## Sales                   1.00000000
```

```
# p-values
corr_result$P
```

```
##                            Store Advertising Costs Size (Sq. Ft)
## Store                         NA      6.961866e-01  7.612028e-01
## Advertising Costs     0.6961866                NA  5.795442e-04
## Size (Sq. Ft)         0.7612028      5.795442e-04            NA
## % Employee Retention  0.4191498      4.421924e-02  4.793537e-03
## Customer Satisfaction 0.4556556      3.164521e-06  1.675365e-06
## Promotion             0.9795360      2.719413e-01  2.340502e-03
## Sales                 0.9446887      5.961867e-05  1.758932e-06
##                      % Employee Retention Customer Satisfaction    Promot
ion
## Store                          0.419149815          4.556556e-01 9.795360e
-01
## Advertising Costs              0.044219240          3.164521e-06 2.719413e
-01
## Size (Sq. Ft)                  0.004793537          1.675365e-06 2.340502e
-03
## % Employee Retention                    NA          2.305167e-02 5.829867e
-01
## Customer Satisfaction          0.023051668                   NA 4.187396e
-01
## Promotion                      0.582986749          4.187396e-01
 NA
## Sales                          0.608816615          9.404499e-06 8.298862e
-06
##                              Sales
## Store                 9.446887e-01
## Advertising Costs     5.961867e-05
## Size (Sq. Ft)         1.758932e-06
## % Employee Retention  6.088166e-01
## Customer Satisfaction 9.404499e-06
## Promotion             8.298862e-06
## Sales                           NA
```

```
## correlation plot
pairs(data_corr[1:7])
```



2.2 - FORWARD SELECTION

```
fitstart <- lm(Sales~1,data = data)
fitstart

##
## Call:
## lm(formula = Sales ~ 1, data = data)
##
## Coefficients:
## (Intercept)
##        1210

full <- lm(Sales ~.,data = data)
fwd <- step(fitstart,
            scope = formula(full),
            direction = "forward")

## Start:  AIC=520.8
## Sales ~ 1
##
##                            Df Sum of Sq      RSS    AIC
## + `Size (Sq. Ft)`           1   5737977  7091222 498.49
## + Promotion                 1   5169224  7659975 501.66
## + `Customer Satisfaction`   1   5121575  7707624 501.91
## + `Advertising Costs`       1   4385811  8443388 505.65
## <none>                                  12829199 520.80
```

```
## + `% Employee Retention`   1     86968 12742231 522.52
## + Store                    1      1604 12827595 522.80
##
## Step:  AIC=498.49
## Sales ~ `Size (Sq. Ft)`
##
##                            Df Sum of Sq     RSS    AIC
## + Promotion                 1   1729572 5361650 489.03
## + `Advertising Costs`       1   1010282 6080940 494.19
## + `Customer Satisfaction`   1    786890 6304331 495.67
## + `% Employee Retention`    1    673165 6418057 496.40
## <none>                                  7091222 498.49
## + Store                     1     24801 7066421 500.35
##
## Step:  AIC=489.03
## Sales ~ `Size (Sq. Ft)` + Promotion
##
##                            Df Sum of Sq     RSS    AIC
## + `Customer Satisfaction`   1   1678748 3682903 475.63
## + `Advertising Costs`       1   1246351 4115299 480.18
## <none>                                  5361650 489.03
## + `% Employee Retention`    1    138181 5223469 489.96
## + Store                     1     13876 5347775 490.92
##
## Step:  AIC=475.63
## Sales ~ `Size (Sq. Ft)` + Promotion + `Customer Satisfaction`
##
##                           Df Sum of Sq     RSS    AIC
## + `Advertising Costs`      1    308276 3374626 474.05
## <none>                                 3682903 475.63
## + `% Employee Retention`   1    136615 3546288 476.08
## + Store                    1     71008 3611894 476.83
##
## Step:  AIC=474.05
## Sales ~ `Size (Sq. Ft)` + Promotion + `Customer Satisfaction` +
##     `Advertising Costs`
##
##                           Df Sum of Sq     RSS    AIC
## + `% Employee Retention`   1    189767 3184859 473.67
## <none>                                 3374626 474.05
## + Store                    1     27669 3346957 475.71
##
## Step:  AIC=473.67
## Sales ~ `Size (Sq. Ft)` + Promotion + `Customer Satisfaction` +
##     `Advertising Costs` + `% Employee Retention`
##
##         Df Sum of Sq     RSS    AIC
## <none>                3184859 473.67
## + Store  1     11055 3173804 475.53
```

```r
summary(fwd)
```

```
##
## Call:
## lm(formula = Sales ~ `Size (Sq. Ft)` + Promotion + `Customer Satisfaction`
 +
##     `Advertising Costs` + `% Employee Retention`, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -752.58  -78.54   33.32  165.38  560.34
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)              -1.762e+03  6.311e+02  -2.792 0.008432 **
## `Size (Sq. Ft)`           2.122e-02  2.052e-02   1.034 0.308304
## Promotion                 5.208e+02  1.187e+02   4.386 0.000101 ***
## `Customer Satisfaction`   4.000e+01  1.446e+01   2.766 0.009005 **
## `Advertising Costs`       4.751e+00  2.384e+00   1.993 0.054107 .
## `% Employee Retention`    8.087e+00  5.600e+00   1.444 0.157602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 301.7 on 35 degrees of freedom
## Multiple R-squared:  0.7517, Adjusted R-squared:  0.7163
## F-statistic:  21.2 on 5 and 35 DF,  p-value: 1.052e-09
```

2.3 - BACKWARD ELIMINATION

```r
back <- step(full,
             direction = "backward")
```

```
## Start:  AIC=475.53
## Sales ~ Store + `Advertising Costs` + `Size (Sq. Ft)` + `% Employee Retent
ion` +
##     `Customer Satisfaction` + Promotion
##
##                            Df Sum of Sq     RSS    AIC
## - Store                     1     11055 3184859 473.67
## - `Size (Sq. Ft)`           1     91525 3265330 474.70
## <none>                                  3173804 475.53
## - `% Employee Retention`    1    173153 3346957 475.71
## - `Advertising Costs`       1    322186 3495991 477.50
## - `Customer Satisfaction`   1    703122 3876927 481.74
## - Promotion                 1   1761184 4934988 491.63
##
## Step:  AIC=473.67
## Sales ~ `Advertising Costs` + `Size (Sq. Ft)` + `% Employee Retention` +
##     `Customer Satisfaction` + Promotion
##
##                            Df Sum of Sq     RSS    AIC
```

```
## - `Size (Sq. Ft)`          1     97258 3282117 472.91
## <none>                                3184859 473.67
## - `% Employee Retention`   1    189767 3374626 474.05
## - `Advertising Costs`      1    361429 3546288 476.08
## - `Customer Satisfaction`  1    696073 3880932 479.78
## - Promotion                1   1750531 4935390 489.63
##
## Step:  AIC=472.91
## Sales ~ `Advertising Costs` + `% Employee Retention` + `Customer Satisfact
ion` +
##     Promotion
##
##                            Df Sum of Sq     RSS    AIC
## - `% Employee Retention`   1    113860 3395978 472.31
## <none>                                3282117 472.91
## - `Advertising Costs`      1    372946 3655063 475.32
## - `Customer Satisfaction`  1   1405913 4688030 485.52
## - Promotion                1   3362431 6644548 499.83
##
## Step:  AIC=472.31
## Sales ~ `Advertising Costs` + `Customer Satisfaction` + Promotion
##
##                            Df Sum of Sq     RSS    AIC
## <none>                                3395978 472.31
## - `Advertising Costs`      1    324381 3720358 474.05
## - `Customer Satisfaction`  1   1300060 4696037 483.59
## - Promotion                1   3660523 7056501 500.29

summary(back)

##
## Call:
## lm(formula = Sales ~ `Advertising Costs` + `Customer Satisfaction` +
##     Promotion, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -754.81 -126.44   -8.95  222.87  495.28
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              -899.824    262.839  -3.423 0.001525 **
## `Advertising Costs`         4.456      2.370   1.880 0.068007 .
## `Customer Satisfaction`    45.130     11.991   3.764 0.000581 ***
## Promotion                 608.785     96.399   6.315 2.35e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 303 on 37 degrees of freedom
```

```
## Multiple R-squared:  0.7353, Adjusted R-squared:  0.7138
## F-statistic: 34.26 on 3 and 37 DF,  p-value: 8.975e-11
```

2.4 - STEP WISE

```
step <- step(full,
             direction = "both")

## Start:  AIC=475.53
## Sales ~ Store + `Advertising Costs` + `Size (Sq. Ft)` + `% Employee Retent
ion` +
##     `Customer Satisfaction` + Promotion
##
##                            Df Sum of Sq     RSS    AIC
## - Store                     1     11055 3184859 473.67
## - `Size (Sq. Ft)`           1     91525 3265330 474.70
## <none>                                  3173804 475.53
## - `% Employee Retention`    1    173153 3346957 475.71
## - `Advertising Costs`       1    322186 3495991 477.50
## - `Customer Satisfaction`   1    703122 3876927 481.74
## - Promotion                 1   1761184 4934988 491.63
##
## Step:  AIC=473.67
## Sales ~ `Advertising Costs` + `Size (Sq. Ft)` + `% Employee Retention` +
##     `Customer Satisfaction` + Promotion
##
##                            Df Sum of Sq     RSS    AIC
## - `Size (Sq. Ft)`           1     97258 3282117 472.91
## <none>                                  3184859 473.67
## - `% Employee Retention`    1    189767 3374626 474.05
## + Store                     1     11055 3173804 475.53
## - `Advertising Costs`       1    361429 3546288 476.08
## - `Customer Satisfaction`   1    696073 3880932 479.78
## - Promotion                 1   1750531 4935390 489.63
##
## Step:  AIC=472.91
## Sales ~ `Advertising Costs` + `% Employee Retention` + `Customer Satisfact
ion` +
##     Promotion
##
##                            Df Sum of Sq     RSS    AIC
## - `% Employee Retention`    1    113860 3395978 472.31
## <none>                                  3282117 472.91
## + `Size (Sq. Ft)`           1     97258 3184859 473.67
## + Store                     1     16788 3265330 474.70
## - `Advertising Costs`       1    372946 3655063 475.32
## - `Customer Satisfaction`   1   1405913 4688030 485.52
## - Promotion                 1   3362431 6644548 499.83
##
## Step:  AIC=472.31
```

```
## Sales ~ `Advertising Costs` + `Customer Satisfaction` + Promotion
##
##                           Df Sum of Sq     RSS    AIC
## <none>                                  3395978 472.31
## + `% Employee Retention`   1    113860 3282117 472.91
## + Store                    1     28743 3367235 473.96
## - `Advertising Costs`      1    324381 3720358 474.05
## + `Size (Sq. Ft)`          1     21351 3374626 474.05
## - `Customer Satisfaction`  1   1300060 4696037 483.59
## - Promotion                1   3660523 7056501 500.29
```

```
summary(step)
```

```
##
## Call:
## lm(formula = Sales ~ `Advertising Costs` + `Customer Satisfaction` +
##     Promotion, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -754.81 -126.44   -8.95  222.87  495.28
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              -899.824    262.839  -3.423 0.001525 **
## `Advertising Costs`         4.456      2.370   1.880 0.068007 .
## `Customer Satisfaction`    45.130     11.991   3.764 0.000581 ***
## Promotion                 608.785     96.399   6.315 2.35e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 303 on 37 degrees of freedom
## Multiple R-squared:  0.7353, Adjusted R-squared:  0.7138
## F-statistic: 34.26 on 3 and 37 DF,  p-value: 8.975e-11
```

3.CONCLUSION

The Adjusted R^2 values obtained from the three variable-selection procedures are:

Model Selection Method	**Adjusted *R2R^2***
Forward Selection	**0.716**
Backward Elimination	**0.713**
Stepwise Selection	**0.713**

The Forward Selection model has the highest adjusted R^2 value of 0.716. This indicates that approximately 71.6% of the variation in Sales is explained by the predictors included in

the final forward-selection model, after adjusting for the number of predictors used in the model.

The Backward Elimination and Stepwise Selection models have an adjusted R^2 of 0.713, indicating that they explain approximately 71.3% of the variation in Sales, after accounting for the number of predictors.

Since the adjusted R^2 of the Forward Selection model (0.716) is slightly higher than those of the Backward and Stepwise models (0.713), the Forward Selection model provides the best fit among the three models based on adjusted R^2. Therefor the best model is

Sales ~ `Size (Sq. Ft)` + Promotion + `Customer Satisfaction` + `Advertising Costs` + `% Employee Retention`.